

RTI Display Controller

Wiring Reference Arduino Mega 2560

This document lists every physical connection needed for the Mega-based RTI display controller: the display itself, the open/close toggle switch, the ignition power relay, the rotary encoder, and the status LEDs. Pair this with the sketch `RTI_control_MEGA.ino` and the project README.

1. RTI Display

The RTI unit is receive-only - it has no data-out/TX line, so it can never send anything back to the Mega. The display communicates over serial at 2400 baud on the Mega's hardware Serial1 port, which keeps USB (Serial, pins 0/1) free for programming and debugging at the same time.

RTI Display Pin	Signal	Mega Pin
Pin 5	RTI RX (data in)	18 (TX1)
Pin 7	GND	GND

Mega pin 19 (RX1) is not used for anything - the RTI unit has no TX line to receive from. Leave it disconnected.

2. Open / Close Toggle Switch

A simple SPDT (or SPST) toggle switch. One leg goes to a digital pin (configured with an internal pull-up), the other leg goes to GND. This switch no longer directly commands the display over serial - it now drives the relay in section 3, which physically cuts or restores the display's ignition power.

Switch Leg	Mega Pin
Signal leg	5
Other leg	GND

Default assumption: pulling the pin LOW (switch closed to GND) means **open**. If your switch behaves backwards, see “Reversed behavior” at the end of this document.

3. Ignition Power Relay

Instead of trying to fake an “off” state over serial (unreliable), the display's actual +12V ignition wire is switched through a relay. Relay ON = ignition power reaches the display = open. Relay OFF = ignition cut = display fully unpowered = closed.

Use a proper relay module (the small board with a built-in transistor driver and flyback diode) - do not wire a bare relay coil directly to a Mega pin, it can't safely supply the current a coil needs.

Connection	Goes To
Relay module signal/IN pin	Mega pin 11
Relay module VCC	Mega 5V (or separate 5V supply if coil draws more current)
Relay module GND	Mega GND
Relay COM	+12V ignition wire
Relay NO (normally open)	Onward to RTI display's ignition input

Wiring uses the relay's NO contacts specifically: de-energized (OFF) leaves the circuit open, so no ignition power reaches the display by default - a safer failure mode than using NC contacts. Most relay modules trigger ON with a HIGH signal, but some cheap active-low boards trigger ON with LOW instead; see “Reversed behavior” if yours is on when it should be off.

4. Rotary Encoder (with pushbutton)

A standard quadrature rotary encoder with an integrated pushbutton (e.g. KY-040 style). CLK must be wired to an interrupt-capable pin - pin 2 on the Mega is used here.

Encoder Pin	Function	Mega Pin
CLK	Quadrature signal A (interrupt)	2
DT	Quadrature signal B	3
SW	Integrated pushbutton	4
GND	Ground	GND
+ / VCC	5V (if your encoder module needs it)	5V

Rotate clockwise to increase brightness, counter-clockwise to decrease. Press SW to cycle video mode (RGB → PAL → NTSC) while the display is open. If direction feels backwards, swap the CLK and DT wires - no code change needed.

5. Status LEDs

Five LEDs give visual feedback on the current state. Wire each through a 220-330Ω current-limiting resistor to GND (standard LED + resistor circuit, anode to the Mega pin, cathode through the resistor to GND).

LED	Behavior	Mega Pin
Status	On when display is open, off when closed	6
Mode: RGB	On when RGB mode is active (and open)	7
Mode: PAL	On when PAL mode is active (and open)	8
Mode: NTSC	On when NTSC mode is active (and open)	9
Brightness	PWM - its own physical brightness mirrors the selected level (0-15 mapped to 0-255)	10

6. Complete Pin Map

Mega Pin	Connected To
0 / 1	Reserved - USB Serial (programming / Serial Monitor)
2	Encoder CLK
3	Encoder DT
4	Encoder SW (pushbutton)
5	Open/Close switch
6	Status LED
7	Mode LED - RGB
8	Mode LED - PAL
9	Mode LED - NTSC
10	Brightness LED (PWM)
11	Ignition relay signal/IN
18 (TX1)	RTI display RX

7. EEPROM Map (for reference, no wiring involved)

Address	Stores
0	Current display mode
1	Current brightness level
2	“Enter” favorite brightness level
3	“Back” favorite brightness level
4	Last video mode used while open

8. Reversed Behavior

These are the most likely “it works, just backwards” issues, and how to fix them without touching the wiring:

- **Switch is backwards** (closed lowers the display instead of raising it): in the sketch, find `checkOpenCloseSwitch()` and change `reading == LOW` to `reading == HIGH`.
- **Relay is backwards** (display is powered when it should be off, or vice versa): change `RELAY_ACTIVE_HIGH` from `true` to `false` near the top of the sketch.
- **Encoder direction is backwards** (clockwise dims instead of brightens): swap the CLK and DT wires at the encoder - no code change needed.
- **One encoder click changes brightness by more than one level**: change `ENCODER_STEPS_PER_CLICK` from 4 to 2 near the top of the sketch.

Note: these wiring assignments and behaviors are based on the sketch as written and standard component conventions - they haven't been verified against your specific physical parts. Confirm each connection against your actual hardware before powering everything on together.

8. Display connector pinout

Control module terminal	Breakout box terminal	Signal type
J1	#1	Color signal (red), screen
J2	#2	Color signal (green), display
J3	#3	Color signal (blue), screen
J4	#4	Synchronizing signal, TV
J5	#5	Screen ground, video signals
J6	#6	Signal, display remote control RX
J7	#7	-
J8	#8	Screen ground, cable to the screen RCA-
J9	#9	Video signal to display RCA+
J10	#10	-
J11	#11	30-supply (power supply from battery) BAT+
J12	#12	30-supply (power supply from battery) BAT+
J13	#13	15 supply ACC (IGNITION)
J14	#14	-
J15	#15	Ground (Measured to battery negative terminal) BAT-
J16	#16	Ground (Measured to battery negative terminal) BAT-
J17	#17	-
J18	#18	Serial bus from the control module TX
J19	#19	-
J20	#20	-
J21	#21	-
J22	#22	-

